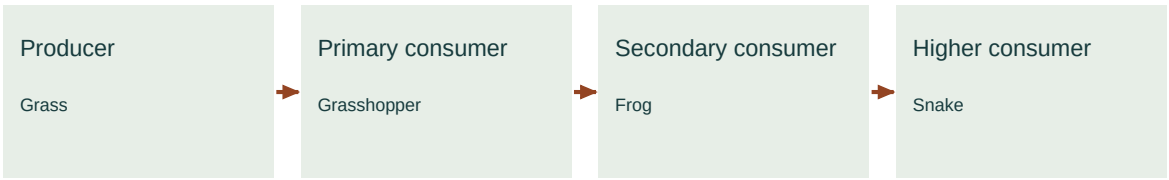
07

Humans and the Environment

A field guide to understanding, working and checking.

Food-chain arrows show transfer to the consumer



Energy transfers along the chain; some is dissipated as heat at each stage.

ENV101 elective: ecosystems, relationships among organisms, pollution, ecological footprint, waste management and sustainable development. The Arabic source lectures are explained here in English.

8 lessons / 2 practice tests / Worked solutions

English edition / November 2026

How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

Your study route

1. Environment and ecosystem components
2. Energy flow, food chains and decomposers
3. Population growth and limiting factors
4. Relationships among organisms
5. Pollution: sources, pathways and effects
6. Ecological footprint and biocapacity
7. Waste management and the hierarchy
8. Sustainable development and integrated decisions

LESSON 01 / 8

Environment and ecosystem components

After this lesson, you can

- Classify living and nonliving ecosystem components.
- Explain one interaction linking an organism with an abiotic factor.

Start with: An ecosystem includes organisms and the physical conditions around them.

THEORY

An ecosystem combines organisms with the physical and chemical surroundings with which they interact. Biotic factors are living components; abiotic factors include light, temperature, water and minerals. A population is one species in an area; a community includes interacting populations.

RULE

Organism → population → community → ecosystem

A method that earns marks

1. Name the ecosystem and define the area being considered.
2. Sort its organisms as biotic and its physical or chemical conditions as abiotic.
3. Choose one organism and explain how a named abiotic factor affects it.
4. Use a causal sentence rather than an unconnected list of examples.

WORKED EXAMPLE

Fish, algae and bacteria are biotic components of a pond. Water temperature and dissolved oxygen are abiotic factors affecting them.

1. Fish, algae and bacteria are organisms, so they are biotic pond components.
2. Water temperature and dissolved oxygen are abiotic conditions even though living things can affect them.
3. Fish depend on sufficient dissolved oxygen for respiration.
4. The example therefore shows both a classification and an interaction between the living community and its physical environment.

Common mistake
A habitat is where an organism lives; an ecosystem includes the interacting system.

Exam habit: For explain questions add an interaction after classifying: name the factor, the organism and the effect.

SELF-CHECK

Are bacteria abiotic because they are microscopic?

Answer: No. Bacteria are living organisms and therefore biotic components.

LESSON 02 / 8

Energy flow, food chains and decomposers

After this lesson, you can

- Assign producer and consumer roles within a stated food chain.
- Distinguish energy flow from nutrient cycling.

Start with: Producers capture energy; consumers obtain energy by feeding; decomposers break down organic material.

THEORY

Producers use light or chemical energy to make organic material from inorganic substances. Consumers obtain organic material by feeding. Decomposers break down dead material and help recycle nutrients. Energy passes through feeding levels and much becomes dispersed heat. Nutrients cycle, but usable energy is not recycled in the same way.

Food-chain arrows show transfer to the consumer



Energy transfers along the chain; some is dissipated as heat at each stage.

Arrows point from food to consumer. This is a simplified food chain; real ecosystems contain food webs. Nutrients cycle, while energy flows through the system.

RULE

Producer → primary consumer → secondary consumer

A method that earns marks

1. Start the chain with its producer and draw arrows in the direction of food-energy transfer.
2. Assign each consumer level from what that organism eats in the stated chain.
3. Explain that energy is used and dissipated at each transfer.

4. Describe decomposers returning nutrients to the environment without claiming energy is recycled.

WORKED EXAMPLE

In grass to grasshopper to frog, grass is the producer, grasshopper the primary consumer and frog the secondary consumer.

1. Grass captures light energy and makes organic material, so it is the producer.
2. The grasshopper eats the producer and is the primary consumer.
3. The frog eats that consumer and is the secondary consumer.
4. The arrows grass → grasshopper → frog show transfer toward the eater; dead material from all levels can enter decomposer pathways.

Common mistake
The 10 percent rule is a rough teaching approximation, not a universal exact constant.

Exam habit: Consumer levels are roles in the specified chain; do not assign a permanent level without looking at the feeding relationship.

SELF-CHECK

Does energy cycle back through the ecosystem in the same way as mineral nutrients?

Answer: No. Energy flows through and is eventually dissipated as heat; nutrients can be recycled.

LESSON 03 / 8

Population growth and limiting factors

After this lesson, you can

- Calculate net population change from four demographic processes.
- Explain how a limiting factor can constrain population growth.

Start with: Population means members of one species in a defined area and period; basic signed addition.

THEORY

Population size changes through births, deaths, immigration and emigration. Rapid exponential growth can occur when resources are abundant. Limited food, space and other conditions constrain growth. Carrying capacity describes a population size an environment can sustain under specified conditions.

RULE

$$\Delta N = \text{births} + \text{immigration} - \text{deaths} - \text{emigration}$$

A method that earns marks

1. Set the population boundary and time interval.
2. Treat births and immigration as additions and deaths and emigration as removals.
3. Add the gains and subtract the losses to obtain net change.
4. If explaining growth limits, link a specific resource or condition to survival or reproduction.

WORKED EXAMPLE

If births are 30, deaths 10, immigration 5 and emigration 8, the population increases by 17.

1. The example gives 30 births and 5 immigrants, so total additions are 35.
2. It gives 10 deaths and 8 emigrants, so total removals are 18.
3. Net change is 35 - 18 = +17 individuals.
4. This is the increase during the interval, not the final population; a starting population would be needed to calculate the latter.

Common mistake
Carrying capacity is not an unchanging property independent of habitat conditions.

Exam habit: Label a positive result increase and a negative result decrease; do not report net change as the final population size.

SELF-CHECK

If births equal deaths, must population size stay unchanged?

Answer: No. Immigration and emigration can still produce a net change.

LESSON 04 / 8

Relationships among organisms

After this lesson, you can

- Identify ecological interactions from their effects on each partner.
- Support an interaction label with scenario evidence.

Start with: Benefit, harm and little measurable effect are relative to the organisms in the interaction.

THEORY

Competition occurs when organisms use a limited shared resource. Predation involves one organism consuming another. In mutualism both benefit; in commensalism one benefits without a substantial effect on the other; in parasitism one benefits at the host expense.

RULE

Mutualism $(+, +)$

commensalism $(+, 0)$

parasitism $(+, -)$

A method that earns marks

1. Identify the two organisms and the resource or activity linking them.
2. Describe the effect on the first organism and on the second separately.
3. Match the two effects to the appropriate interaction category.
4. Use evidence from the case rather than assuming all close associations are mutualistic.

WORKED EXAMPLE

A parasite obtaining nutrients while harming its host is a plus-minus interaction. Two plants competing for limited light are both constrained by competition.

1. The parasite obtains nutrients, so it benefits from the interaction.
2. The host is harmed, giving the benefit/harm pattern commonly written plus/minus.
3. The two competing plants both face restricted access to limited light.
4. Competition is therefore a minus/minus interaction in the comparison with unrestricted resource access, even if one competitor does better than the other.

Common mistake
A relationship may be context-dependent; answer using the effects given in the question.

Exam habit: Write the effect on each partner before naming the interaction; this makes similar scenarios easier to distinguish.

SELF-CHECK

Does one competitor winning mean competition is automatically a benefit/harm interaction?

Answer: No. Both incur the effects of competing for a limited resource; one may still outperform the other.

LESSON 05 / 8

Pollution: sources, pathways and effects

After this lesson, you can

- Build a source-pathway-receptor explanation of pollution harm.
- Select prevention that interrupts a stated nutrient-runoff pathway.

Start with: Nutrients support growth; decomposition can consume dissolved oxygen.

THEORY

Pollution changes environmental conditions in ways that can cause harm or reduce usefulness. Sources can be natural or human-made. Air, water and soil pollution are connected through transport and deposition. A useful answer links a source to a pathway, an effect and a control.

RULE

Source → pathway → receptor → effect

A method that earns marks

1. Name the pollution source and the released material.
2. Trace how the material reaches the affected ecosystem.
3. Explain the intermediate process connecting exposure to the final effect.
4. Choose a preventive measure and state which link it reduces.

WORKED EXAMPLE

Nutrient-rich runoff can stimulate excessive algal growth. Subsequent decomposition can consume dissolved oxygen and harm aquatic organisms.

1. Excess nutrients from fertilizer can be carried by runoff into a water body.
2. Nutrient enrichment can stimulate excessive algal growth.
3. When organic material is decomposed, microbial respiration can consume dissolved oxygen.
4. Low oxygen can harm fish; reducing excess fertilizer or intercepting runoff addresses the source or pathway before that effect occurs.

Common mistake
Do not say algae always produce an immediate oxygen shortage without explaining decomposition and conditions.

Exam habit: Use connecting verbs such as carries, stimulates and consumes to make every step in the causal chain explicit.

SELF-CHECK

Why is algae cause fish death an incomplete explanation?

Answer: It omits the conditions and intermediate processes, especially decomposition and dissolved-oxygen depletion.

LESSON 06 / 8

Ecological footprint and biocapacity

After this lesson, you can

- Compare ecological demand with biocapacity on a common basis.
- Interpret a numerical balance as a deficit or reserve.

Start with: Demand versus available capacity; subtraction; per-person and total quantities must not be mixed.

THEORY

Ecological footprint estimates demand for biologically productive area needed to support consumption and absorb certain wastes. Biocapacity represents available biological production capacity. A deficit occurs when demand exceeds capacity. A global hectare is a productivity-adjusted unit, not simply any local hectare.

RULE

$$\text{Ecological deficit} = \text{footprint} - \text{biocapacity}$$

if positive

A method that earns marks

1. Identify footprint as demand and biocapacity as available biological production capacity.
2. Check that both figures use the same units, time basis and population basis.
3. Subtract biocapacity from footprint to test for a deficit.
4. Explain the result within the indicator scope rather than treating it as every environmental impact.

WORKED EXAMPLE

A population with footprint 3 global hectares per person and biocapacity 2 has a deficit of 1 per person.

1. The example footprint is 3 global hectares per person, describing demand.
2. Biocapacity is 2 global hectares per person, describing available capacity on the same basis.
3. Demand exceeds capacity by 3 - 2 = 1 global hectare per person.
4. The result is an ecological deficit, not an area measured in ordinary unadjusted local hectares.

Common mistake
Ecological footprint and carbon footprint are related but not identical measures.

Exam habit: Write global hectares per person in a per-person result; omitting the population basis changes its meaning.

SELF-CHECK

Can a region total footprint be compared directly with biocapacity per person?

Answer: No. Convert both to totals or both to per-person values first.

LESSON 07 / 8

Waste management and the hierarchy

After this lesson, you can

- Distinguish prevention, reuse, recycling, recovery and disposal.
- Apply the waste hierarchy to a concrete material scenario.

Start with: Reuse keeps an item in service; recycling reprocesses its material.

THEORY

Waste management involves preventing, collecting, sorting, treating and disposing of waste. Prevention and reduction usually come before reuse, recycling, recovery and final disposal. Hazardous items require appropriate separate handling; not every household discard belongs in an ordinary bin.

RULE

Prevent/reduce → reuse → recycle → recover → dispose

A method that earns marks

1. Identify whether the action avoids waste, keeps the item in use or processes discarded material.
2. Classify the action using the general hierarchy.
3. For a proposed choice consider prevention or reduction first where feasible.
4. Check material hazards and the approved local collection stream before actual disposal.

WORKED EXAMPLE

Refilling a reusable bottle is reuse. Turning sorted used glass into new glass is recycling. Avoiding unnecessary packaging is prevention.

1. Refilling a durable bottle keeps the same item in service, so it is reuse.
2. Processing separated used glass into new glass changes the material into a new product, so it is recycling.
3. Avoiding unnecessary packaging prevents that material becoming waste at all.
4. These actions belong at different stages of the hierarchy and should not all be labelled recycling.

Common mistake
Recycling does not erase all upstream resource use or justify unlimited consumption.

Exam habit: Use the action in the scenario as evidence: avoid means prevention, use again means reuse, reprocess means recycling.

SELF-CHECK

Is using a clean jar again to store suitable items recycling?

Answer: It is reuse, because the same item is used again without material reprocessing.

LESSON 08 / 8

Sustainable development and integrated decisions

After this lesson, you can

- Assess a project across environmental, social and economic dimensions.
- State measurable indicators and a trade-off for a sustainability decision.

Start with: Present and future needs; a benefit in one dimension can have costs in another.

THEORY

Sustainable development addresses present needs while preserving future ability to meet needs. Environmental, social and economic dimensions interact. A project can improve one dimension while creating costs in another, so evaluation should consider benefits, trade-offs and long-term effects.

RULE

Environment + society + economy

A method that earns marks

1. Define the project and the needs it is intended to meet.
2. Give one environmental, one social and one economic effect.
3. Select an indicator that could test each effect.
4. Explain one trade-off over the project life rather than declaring the project sustainable from a single benefit.

WORKED EXAMPLE

A public-transport project can reduce emissions, improve access and lower travel costs, but its construction and operation also require resources and funding.

1. The public-transport example can reduce transport emissions if its use changes travel patterns.
2. Improved access to education or work is a social benefit to evaluate.
3. Travel costs, operating costs and funding provide economic questions.
4. Construction also consumes resources and money, so a full assessment compares these costs with benefits over time.

Common mistake
Do not equate sustainable development with only economic growth or only nature protection.

Exam habit: Replace vague words such as greener with a measurable indicator such as electricity use, emissions, access or cost over a stated period.

SELF-CHECK

Is lower operating energy alone enough to prove a project is sustainable?

Answer: No. Social access, economic viability, construction impacts and long-term effects also require assessment.

Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

Humans and the Environment / Test A

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which is an abiotic factor?

   A. Algae

   B. Bacteria

   C. Water temperature

   D. Fish

2. Decomposers mainly help by:

   A. Recycling nutrients from dead matter

   B. Creating sunlight

   C. Stopping all energy loss

   D. Removing all producers

3. Which directly reduces a local population?

   A. Reproduction

   B. Births

   C. Immigration

   D. Emigration

4. Both organisms benefit in:

   A. Parasitism

   B. Mutualism

   C. Predation

   D. Competition

Part II: Show your reasoning

1. Classify sunlight, grass, soil mineral content and a rabbit as biotic or abiotic.

2. A question specifies 10 percent transfer efficiency at each step. Producers have 10000 energy units. Find energy available to primary and secondary consumers.

3. A population begins with 200 individuals. During a period there are 25 births, 12 deaths, 8 immigrants and 6 emigrants. Find the final size.

4. Identify a relationship where both benefit and another where one benefits while the other is harmed. Explain the signs.

Humans and the Environment / Test B

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Excess nutrient input into water is most closely associated with:

   A. Eutrophication

   B. Lens refraction

   C. Sterilization

   D. No biological change

2. A footprint larger than biocapacity indicates:

   A. Unlimited sustainability

   B. An ecological reserve

   C. An ecological deficit

   D. Zero resource use

3. Which is highest in the general waste hierarchy?

   A. Disposal after mixing

   B. Landfill

   C. Uncontrolled burning

   D. Waste prevention

4. A complete sustainability assessment considers:

   A. Only this month

   B. Environmental, social and economic effects

   C. Only profit

   D. Only tree numbers

Part II: Show your reasoning

1. Explain a three-step route from excess fertilizer use to fish harm and name one preventive measure.

2. A region has footprint 4.5 and biocapacity 2.8 global hectares per person. Identify and calculate the balance.

3. Classify avoiding excess packaging, reusing a jar, reprocessing aluminium cans and placing residual waste in an approved landfill.

4. Give one environmental, one social and one economic indicator for a campus energy-efficiency project, plus one trade-off to evaluate.

Humans and the Environment / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **C.** Temperature is a physical condition rather than a living organism.
2. **A.** They break down organic matter and return nutrients to the system.
3. **D.** Emigration removes individuals from the local population.
4. **B.** Mutualism is defined by benefits to both participants.

Part II: Written solutions

1. Sunlight and mineral content are abiotic. Grass and rabbit are biotic. One mark each.
2. Primary consumers receive 1000 units; secondary consumers receive 100 units. Two marks for each transfer. These values depend on the stated efficiency.
3. Net change =25-12+8-6=15. Final size=215. Credit signs of all four processes and final total.
4. Mutualism is (+,+): both gain a benefit. Parasitism is (+,-): parasite benefits and host is harmed. One mark for each name and each explanation.

Humans and the Environment / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **A.** Nutrient enrichment can drive excessive growth and subsequent oxygen depletion.
2. **C.** Demand exceeds the available biological capacity under the accounting model.
3. **D.** Avoiding waste reduces the need for later treatment and disposal.
4. **B.** The concept integrates all three dimensions and future consequences.

Part II: Written solutions

1. Runoff carries nutrients to water; algal growth increases; decomposition can lower dissolved oxygen and harm fish. Reduce excess nutrient application or intercept runoff. One mark for each causal step and one for prevention.
2. It has an ecological deficit of 4.5-2.8=1.7 global hectares per person. Credit comparison, subtraction, units and deficit interpretation.
3. Prevention, reuse, recycling and disposal, respectively. One mark for each classification.
4. Examples: electricity use/emissions; comfort or accessibility; cost savings/payback; upfront cost versus long-term savings. One mark for each relevant, measurable item.

Source and coverage notes

ENV101 elective: ecosystems, relationships among organisms, pollution, ecological footprint, waste management and sustainable development. The Arabic source lectures are explained here in English.

Source folder: Open original course folder The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. This revised study aid does not replace lecturer-specific derivations or instructions. All practice tests and suggested marking guidance are original study material.